

Identifying Financially Constrained Firms: Methodology

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Outline

Setup and the question

Diff-in-diff

Triple diff

Challenges in measuring the return to the shock

Capital demand turns on opportunity and the wedge

Firm's FOC (profit max with a financing wedge $\tau_i \geq 0$):

$$MRPK_{i,t} = \rho_{i,t} = (1 + \tau_{i,t}) r_{i,t}^{adj}$$

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$MRPK$ falls in $K \Rightarrow$ invert the FOC to get capital demand [details]

$$\Delta \log K_{i,t} = \varepsilon_{\Theta} \Delta \log \Theta_{i,t} - \varepsilon_K \Delta \log \rho_{i,t}$$

Θ = investment opportunity τ = financing wedge r^{adj} = risk-adjusted cost (the welfare benchmark)

$$\varepsilon_{\Theta} = \varepsilon_K = \frac{1}{1-\alpha}$$

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$\rightarrow$ We want $\{\tau_i\}$, but it is buried inside $\rho = (1 + \tau) r^{adj}$

How to identify financing frictions

Two tools in the empirical literature:

- Diff-in-diff (this section)
- Triple diff (the cross-section of *MRPK*)

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🎯 Lecture objective

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The DiD shocks financing but cannot say what it moved

Shock financing directly, read the **average** response (e.g. Chodorow-Reich 2014; Banerjee-Duflo 2014)

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$$S_i = -\varepsilon_K \Delta \log \rho_i + \varepsilon_\Theta \Delta \log \Theta_i$$

(A) Orthogonality: if $\text{Cov}(S_i, \Delta \log \Theta_i) \neq 0$, $\hat{\beta}$ blends **opportunity** with the **cost** channel. Orthogonality is an **assumption**, not a test

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$$\Delta \log \rho_i = \Delta \log r_i^{adj} + \Delta \log(1 + \tau_i)$$

(B) Joint identification: $\Delta \rho$ splits into the fundamental r^{adj} and the wedge $\tau \rightarrow \hat{\beta}$ cannot separate

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(B) Joint identification: $\Delta \rho$ splits into the fundamental r^{adj} and the **wedge** $\tau \rightarrow \hat{\beta}$ **cannot separate**

Key takeaway

One number **two** objects to identify $\rightarrow$ **pb** $\rightarrow$ Need to add the **cross-section**

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Challenges in measuring the return to the shock

The friction shows up in the interaction, not the average effect

Split the shock S_i by ex-ante $MRPK$. $FC_i = 1$ for ex-ante high- $MRPK$ firms (financing-constraint proxy):

$$\Delta \log K_i = \beta_2 S_i + \beta_1 (FC_i \times S_i) + [FE]$$

- β_2 : how much low- $MRPK$ firms respond to the shock
- β_1 : how much *more* high- $MRPK$ firms respond
- $\beta_2 + \beta_1$: how much shocked high- $MRPK$ firms respond relative to non-shocked high- $MRPK$

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Theory can give us a **sign identifying moment**: β_1 different if:

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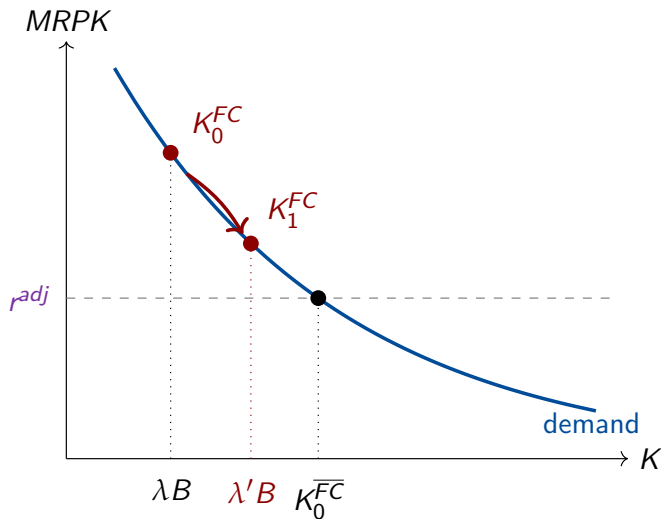
But high $MRPK$ can still mean **constrained** or just **risky**

Theory can give us a **sign identifying moment**: β_1 different if:

- S_i = shock to user cost $\rho = (1 + \tau) \rho^{adj}$
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→ **Identification question**: did it move the **friction** τ or the fundamental ρ^{adj} ?

Shock the friction τ : the rationed firm jumps



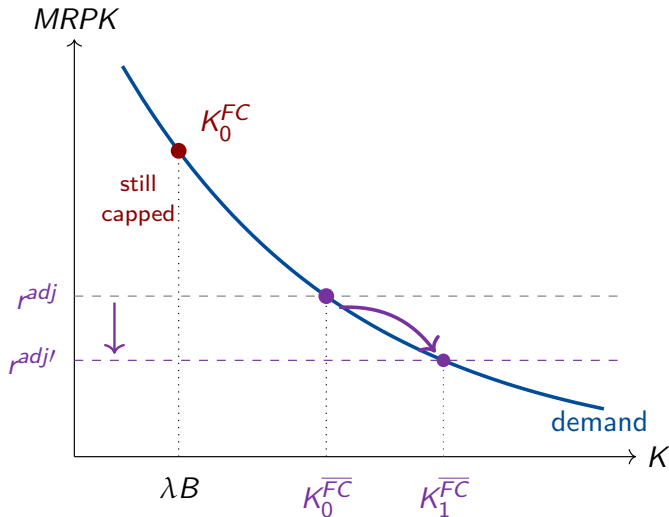
Relax the friction (cap $\lambda B \rightarrow \lambda' B$):

- the rationed **high-MRPK** firm K_0^{FC} jumps toward its optimum, $MRPK \rightarrow r^{adj}$ (reaching K_1^{FC})
- the unconstrained **low-MRPK** firm sits at $K_0^{\overline{FC}}$ and does not move

$\Rightarrow$ high-MRPK reacts more

$\Rightarrow \beta_1 > 0$

Shock the fundamental r^{adj} : the unconstrained firm moves



Lower the fundamental ($r^{adj} \rightarrow r^{adj'}$):

- the unconstrained **low-MRPK** firm slides down its curve from $K_0^{\overline{FC}}$ to $K_1^{\overline{FC}}$ (K rises)
- the rationed **high-MRPK** firm K_0^{FC} is still capped at λB

$\Rightarrow$ low-MRPK reacts more

$\Rightarrow \beta_2 > 0$

$\Rightarrow \beta_1 < 0$

Recap: the sign of β_1 tells τ from r^{adj}

$$\Delta \log K_i = \beta_2 S_i + \beta_1 (\text{HighMRPK}_i \times S_i) + [\text{FE}]$$

- shock the friction τ (relief): **high-MRPK** jumps $\Rightarrow \beta_1 > 0$
- shock the fundamental r^{adj} : **low-MRPK** moves $\Rightarrow \beta_2 > 0$

Key takeaway

- the *sign* of β_2 says whether the shock moved the **friction** or the **fundamental**
- the *sign* of β_1 says whether the friction is a **quantity** or **price premium**

Caveat: clean only if the shock does not reprice **risk** differentially by **MRPK** [details] $\Rightarrow$ we need to **also shock** ☹
(next slide)

Alternative method: Shock on Θ

Hold the cost; shock opportunity, read the cross-section of constraint (Rajan-Zingales 1998)

$$\Delta \log K_i = \varepsilon_0 \Delta \log \Theta_i + \varepsilon_1 (FC_i \times \Delta \log \Theta_i) + \dots$$

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Key takeaway

A high-MRPK firm that ignores opportunity (Θ) is **quantity constrained**, not **risky**

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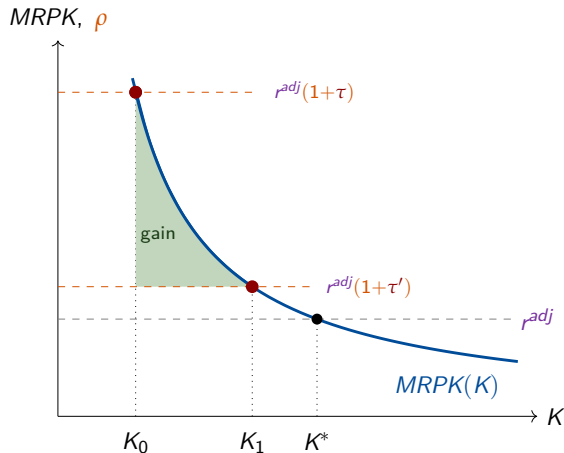
Diff-in-diff

Triple diff

Challenges in measuring the return to the shock

The natural thing: measure the welfare gain from relief

Relief adds capital. Each extra unit is worth $MRPK - r^{adj} > 0$



Quantity-constrained firm: the cap acts like a wedge, so it sits where

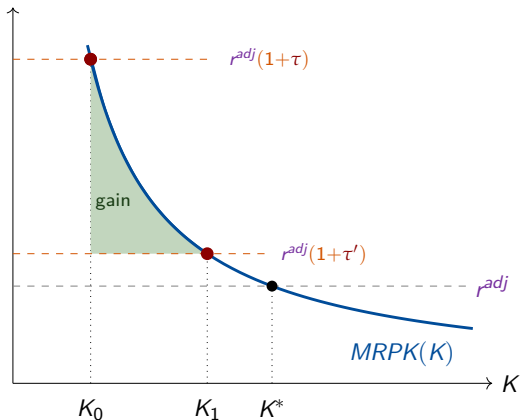
$$MRPK = r^{adj}(1 + \tau)$$

- relief $\tau \downarrow$: K rises $K_0 \rightarrow K_1$
- Extra investment is $NPV > 0 = \int_{K_0}^{K_1} (MRPK - r^{adj}(1 + \tau')) dK$

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$MRPK, \rho$



Can we just estimate the friction from:

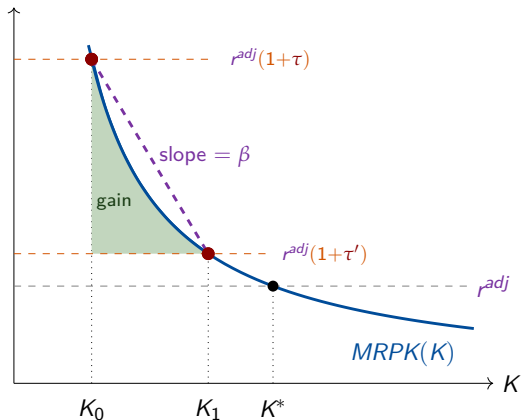
$$\Delta \log MRPK_i = \beta \Delta \log K_i + \alpha_i + \delta_t,$$
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with $\beta > 0$?

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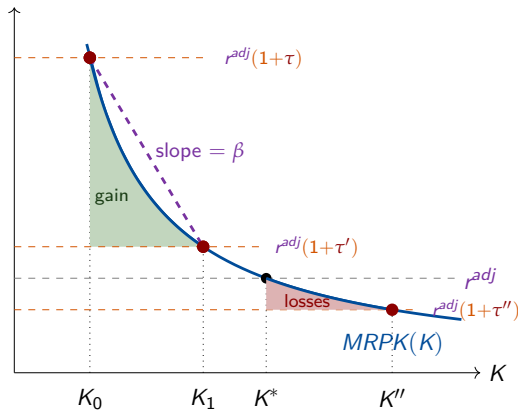
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No (1): β measures the curvature of the technology, not the size of the friction

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No (2): $\Delta K > 0$ can be **bad**

More investment helps only while $MRPK > r^{adj}$

The welfare object is a **sign**, and $\Delta K > 0$ can be **bad**. Marginal social value of the firm's capital:

$$\frac{dV_i}{dK_i} = MRPK_i - r^{adj} = \tau_i r^{adj}$$

- $\tau > 0$ (underinvest, $MRPK > r^{adj}$): more K is **good**
- $\tau = 0$ first best ; $\tau < 0$ (overinvest, $MRPK < r^{adj}$): more K is **bad**

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Two **cases** where the DiD's $\Delta K > 0$ is **bad** (**overinvestment**):

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- Risk subsidy: financing below the true $r^{adj} \Rightarrow \tau < 0 \Rightarrow$ **overinvestment**
- Empire-building: $MRPK < r^{adj}$ even at $\tau = 0$; the collateral constraint was **disciplining**, so relaxing it $\Rightarrow$ **overinvestment** [details]

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Key takeaway

More capital is **not** always **welfare improving**

What to remember

Key takeaway

A difference-in-differences shows that capital **moves**. Only a **triple difference** can tell whether that movement **repairs a financing friction** or merely reflects **risk** or a **subsidy**

- Welfare hinges on the **sign of $MRPK - r^{adj}$** , not the size of ΔK
- The **ρ** shock sign-identifies; the **$\ominus$** shock validates the proxy
- Different failure modes, so agreement across them is the evidence

Thank you!

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Appendix

From FOC to capital demand

Step 1. FOC, with $MRPK = \tilde{\alpha} R/K$ (R = revenue):

$$\tilde{\alpha} \frac{R_i}{K_i} = \rho_i \implies \Delta \log K_i = \Delta \log R_i - \Delta \log \rho_i \quad (\text{conditional: } R \text{ fixed})$$

Step 2. But R_i is endogenous. Downward-sloping demand $\Rightarrow R_i = \Theta_i K_i^{\tilde{\alpha}}$:

$$\Delta \log R_i = \Delta \log \Theta_i + \tilde{\alpha} \Delta \log K_i$$

Step 3. Substitute out R_i and collect:

$$\Delta \log K_i = \underbrace{\frac{1}{1-\tilde{\alpha}} \Delta \log \Theta_i}_{\varepsilon_{\Theta}} - \underbrace{\frac{1}{1-\tilde{\alpha}} \Delta \log \rho_i}_{\varepsilon_K}$$

$\Theta_i = \underbrace{D_i^{1/\sigma}}_{\text{demand}} \underbrace{A_i^{(\sigma-1)/\sigma}}_{\text{productivity}}$: investment opportunity; TFP is the special case.

[back]

Capital demand: two margins (detail)

$$\Delta \log K_i = \varepsilon_{\Theta} \Delta \log \Theta_i - \varepsilon_K \Delta \log \rho_i, \quad \rho_i = (1 + \tau_i) r_i^{adj}$$

- Two RHS margins: investment opportunity Θ and the user cost $\rho = (1 + \tau) r^{adj}$
- We want $\{\tau_i\}$, but it is buried inside ρ
- Every method = move one margin, read the other from the cross-section

[back]

The wedge τ_i has two faces (formal)

- Price face: $\rho_i = (1 + \tau_i) r_i^{adj}$ raises the user cost; shifts the *level* of K down; a fixed proportional wedge leaves ε_Θ unchanged
- Quantity face: τ_i = multiplier on a collateral limit $K_i \leq \lambda_i B_i$; slack $\Rightarrow \tau_i = 0$ (on the curve), binding $\Rightarrow$ pinned at $\lambda_i B_i$ (off the curve)

Same object links them: $\tau_i = \frac{MRPK_i}{r_i^{adj}} - 1$

The quantity face is load-bearing: under a pure proportional price wedge ε_Θ is common across firms and the opportunity-shock test has no bite

Empire-building and the welfare integral

Empire-building. Manager values size, $B(K)$ with $B' > 0$. Manager FOC:

$$R'(K) + B'(K) = (1 + \tau) r^{adj} \implies MRPK - r^{adj} = \tau r^{adj} - B'(K)$$

At $\tau = 0$ this is $-B'(K) < 0$: the firm **overinvests** absent any wedge, so the collateral constraint is **disciplining**; relaxing it lowers welfare

Welfare integral. To first order,

$$\Delta W \approx \sum_i (MRPK_i - r^{adj}) \Delta K_i = \sum_i \tau_i r^{adj} \Delta K_i,$$

so the profitability-by- ΔK comovement is the welfare test

[back]

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The treatment S_i is common across $MRPK$, but β_1 is the differential effect, and a common shock can pass through differentially

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Foothold: if high- $MRPK$ firms are risky, their risk premium can fall ~ 2 pp more when financing is eased $\Rightarrow a_1 < 0$, $\beta_1 > 0$, with no friction. We wish $a_1 = 0$; nothing in the design delivers it

→ clean only if $a_1 = 0$ and the shock carries no differential opportunity ($\Delta \Theta$) by $MRPK$